Yahya Jabary
yjabary@ethz.ch | +41 78 221 22 60
Swiss B-Permit, EU Citizen (Austrian) | Native German, English fluent
github.com/sueszli | scholar.google.com/SH4-xwEAAAAJ



Education

ETH Zurich	Master's Thesis	Machine Learning	*Jul 2024 - Mar 2025*
TU Wien	Dipl.-Ing.	Computer Science	*Sep 2019 - Mar 2025*

Experience

ETH Zurich — Zurich, Switzerland
Research Assistant — *Jul 2026 - present*
- Received a Ph.D. offer.
- Studying self-improving task generation for robot learning through physical puzzles.

Python PyTorch MuJoCo

University of Cambridge — Cambridge, UK
Research Assistant — *Dec 2025 - May 2026*
- Wrote a custom JIT compiler for MicroGPT, reaching 5.3x over PyTorch and 14.4x over JAX.
- Landed 130+ commits across xDSL's LLVM backend and a formally verified IR in Lean.
- Recommendation Letter: sueszli.github.io/rec-cam.pdf

Python Lean MLIR LLVM

Qube Research & Technologies — Paris, France
Quantitative Technology Analyst, Internship — *Apr 2025 - Oct 2025*
- Designed an ensemble of statistical volatility indicators per asset class.
- Ported hot paths to zero-copy CPython extensions in Rust via PyO3, reaching 2.1x speedup.
- Validated the port against terabytes of numerical data with a differential fuzzing harness.

Python Rust Apache Arrow

ETH Zurich — Zurich, Switzerland
Research Assistant — *Jul 2024 - Mar 2025*
- Reproduced DeepMind's self-ensemble paper, tracing robustness to training, not architecture.
- First-authored the paper below on simple geometric attacks that transfer across CNNs and ViTs.
- Recommendation Letter: sueszli.github.io/rec-eth.pdf

Python PyTorch JAX

University of Waterloo — Ontario, Canada
Research Assistant — *Nov 2023 - Jun 2024*
- Received a direct-entry Ph.D. offer from my Bachelor's.
- Synthesized the first code-embedding dataset by modifying ASTs and benchmarked vector databases.

Python PyTorch JAX

Erste Asset Management — Vienna, Austria
Trading Software Engineer, Internship — *Apr 2023 - Oct 2023*
- Offered a permanent position.
- Rewrote a multi-process Java FIX gateway as a single in-memory service, removing IPC.

Java Terraform FIX Protocol

Publications

Jabary, Y. (2025). Rethinking Adversarial Examples (Master's Thesis, ETH Zurich). [code, pdf]

Jabary, Y., Plesner, A., Kuzhagaliyev, T., & Wattenhofer, R. (2024). Seeing Through the Mask: Rethinking Adversarial Examples for CAPTCHAs. [news, code, pdf]

Honors & Awards

ETH AI Center, Associated Researcher	*2026*
University of Cambridge, Formal Reasoning Summer School, Full Scholarship (£1,200)	*2026*
ACM Europe, MLIR Summer School, Full Scholarship (€550)	*2026*
Qube Research & Technologies, Hackathon Judge	*2025*
Google for Developers, Chapter Founder	*2024*
EPFL, Summer Research Institute SuRI Fellowship	*2024*
University of Waterloo, Graduate Research Studentship GRS ($23,081)	*2023*
University of Waterloo, Mathematics International Master's Award of Excellence M-IMAE ($12,500)	*2023*
TU Wien, Institute of Logic & Computation Research Scholarship (€1,200)	*2021*
Austrian Government, Education Grant (€3,000)	*2018*
University of Vienna, Gifted Students Early Admission	*2017*